

The empirical recurrence for OEIS A254722, recovered from its tail

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Abstract

OEIS A254722 records “Empirical recurrence of order 75” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 295$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 17$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{75} = -768 \neq 0$, so no further tail satisfies anything shorter; any order-75 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254722 is “Number of $(n+2) \times (2+2)$ 0..1 arrays with every 3×3 subblock sum of the two maximums of the central row and the central column and the two medians of the diagonal and antidiagonal nondecreasing horizontally and vertically.”. It has offset 1 and begins

2630, 14226, 76615, 364039, 1645201, 7620738, 35110289, 160937385, ...

The entry states:

Empirical recurrence of order 75 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:53:23, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 295$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 295$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 75: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 17$ is the first whose minimal order is exactly the stated 75.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 590$ exact terms from $n = 17$ on, it returns order 75 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	8	c_{20}	−19313	c_{39}	358780	c_{58}	−131491
c_2	−15	c_{21}	−24986	c_{40}	−501897	c_{59}	470249
c_3	0	c_{22}	−28214	c_{41}	65972	c_{60}	−287386
c_4	−27	c_{23}	2211	c_{42}	131758	c_{61}	119349
c_5	−28	c_{24}	37022	c_{43}	−270660	c_{62}	35964
c_6	557	c_{25}	−36721	c_{44}	346196	c_{63}	−252697
c_7	−534	c_{26}	63615	c_{45}	−26095	c_{64}	207524
c_8	−135	c_{27}	−98921	c_{46}	179006	c_{65}	−90161
c_9	−339	c_{28}	9510	c_{47}	−106652	c_{66}	10158
c_{10}	−3470	c_{29}	169864	c_{48}	−125971	c_{67}	73689
c_{11}	7112	c_{30}	−106502	c_{49}	−69754	c_{68}	−73652
c_{12}	1234	c_{31}	160711	c_{50}	−356605	c_{69}	36356
c_{13}	−1152	c_{32}	−176401	c_{51}	464129	c_{70}	−12884
c_{14}	7186	c_{33}	−204059	c_{52}	−32568	c_{71}	−6532
c_{15}	−29622	c_{34}	186911	c_{53}	103678	c_{72}	10264
c_{16}	1309	c_{35}	−240283	c_{54}	276575	c_{73}	−5728
c_{17}	14336	c_{36}	409214	c_{55}	−576689	c_{74}	2880
c_{18}	699	c_{37}	60598	c_{56}	195078	c_{75}	−768
c_{19}	46477	c_{38}	−256128	c_{57}	−111138		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A254722 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 17$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{75} - c_1 x^{75-1} - \dots - c_{75}$. Its constant term is $-c_{75} = 768$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 75 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 75, so $\deg q = 75$, and it is monic. It holds from some index t on. From $\max(t, 17)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 75, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 75, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -768 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-75 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A254722.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.